Axioms, Conditioning & Bayes: 10 Hard Problems

Full Chapter 1.2 Coverage, Jane Street Stage-1 Calibration — Every Question Fuses Two or More Ideas

Same philosophy as the combinatorics hard set: nothing here is solved by one formula in isolation. Several of these are direct escalations of problems from your base 20-problem sheet — H6 escalates Monty Hall, H7 escalates Bertrand's Box, H8 escalates the Boy-or-Girl paradox, and H9 is a deliberate, worked example of *why* the sequential-odds trick from Problem 15 can quietly fail in the real world. Every option set has exactly one correct answer and three answers built from a specific, common error.

I. Kolmogorov Framework & Event Spaces

Problem H1. *[Technique: Chained Bonferroni bound for a triple intersection]*

Events A, B, C satisfy $P(A) = 0.8$, $P(B) = 0.75$, $P(C) = 0.7$ (no independence assumed). Using only the axioms, find the tightest lower bound the axioms alone can guarantee on $P(A \cap B \cap C)$.

Options: (A) 0 (B) 0.25 (C) 0.55 (D) 0.7

Solution. Apply the two-event bound twice. First, $P(A \cap B) \geq P(A) + P(B) - 1 = 0.8 + 0.75 - 1 = 0.55$. Now treat $A \cap B$ as a single event and apply the bound again against C:

$$P((A \cap B) \cap C) \geq P(A \cap B) + P(C) - 1 \geq (0.55) + 0.7 - 1 = \boxed{0.25}.$$

In general, $P(A \cap B \cap C) \geq P(A) + P(B) + P(C) - 2$ (chained Bonferroni). Here $0.8 + 0.75 + 0.7 - 2 = 0.25 > 0$, so this is a genuine, non-vacuous constraint.

Trap analysis: 0.55 = stopped after only the first (two-event) application, forgetting to fold in C. 0.7 = $\min(P(A), P(B), P(C))$ — a true fact about the *upper* bound on the intersection ($P(A \cap B \cap C) \leq \min_i P(A_i)$ always), but the question asked for the *lower* bound, and confusing the two directions is a very natural mistake under time pressure. 0 = correctly recalling that the raw formula can go negative (in which case you'd clip to 0) but failing to actually compute the formula here, where it stays safely positive.

★ *Intuition:* Each application of the two-event bound is like squeezing a fixed 100%-budget pie between two overlapping claims; chaining it twice is squeezing the same pie between three claims in sequence — the "slack" you're allowed shrinks by 100 percentage points with every event you add, which is exactly why three fairly large probabilities can already force a strictly positive triple overlap.

Correct: (B) 0.25

Problem H2. *[Technique: Boole's inequality applied to the matching problem (links Ch. 1.1 and 1.2)]*

A shuffled 52-card deck has $A_i = \{\text{card } i \text{ is in its original position}\}$, $i = 1, \ldots, 52$, each with $P(A_i) = \frac{1}{52}$. What does Boole's inequality give as an upper bound on $P(\text{at least one match})$, and is this bound useful?

Options: (A) $\frac{1}{52}$ (B) $1 - \frac{1}{e} \approx 0.632$ (C) 1 (exactly, vacuous) (D) $\frac{52}{e}$

Solution. Boole: $P(\bigcup A_i) \leq \sum_{i=1}^{52} P(A_i) = 52 \times \frac{1}{52} = \boxed{1}$. This is exactly the trivial bound "≤ 1" — completely vacuous. The true answer (from the derangement/matching-problem machinery of Chapter 1.1) is $P(\text{at least one match}) = 1 - D_{52}/52! \approx 1 - \frac{1}{e} \approx 0.632$.

Trap analysis: $1 - 1/e \approx 0.632$ is the *true* value, not Boole's bound — conflating "the bound" with "the answer" is the trap. $\frac{1}{52}$ = forgot to sum over all 52 events, reporting a single term instead. $\frac{52}{e}$ is a nonsense mangling of the derangement approximation $D_n \approx n!/e$ with the union-bound machinery.

★ *Intuition:* This is the deepest possible illustration of "Boole's bound is only useful when $n \times p \ll 1$" from your base sheet: here $n \times p = 52 \times \frac{1}{52} = 1$ *exactly*, by construction (the expected number of matches in a random permutation is always exactly 1, for any n, via the indicator-variable argument from Notebook 1). So the union bound sits precisely on the boundary between "informative" and "vacuous" for every deck size, not just 52 — a beautiful, non-accidental fact.

▶ *Speed insight:* Whenever you recognize a "match/fixed-point" structure, immediately recall $\mathbb{E}[\text{matches}] = 1$ regardless of n — this instantly tells you Boole's bound will always be exactly 1 (vacuous) for this entire family of problems, no computation needed.

Correct: (C) 1 (exactly, vacuous)

Problem H3. *[Technique: Solving for the exact fairness condition behind pairwise independence]*

Two coins, each with $P(\text{heads}) = p$ (independent of each other), are flipped. Let $A = \{\text{coin 1 heads}\}$, $B = \{\text{coin 2 heads}\}$, $C = \{\text{coins match}\}$. For which value(s) of $p \in (0, 1)$ are A, B, C pairwise independent?

Options: (A) $p = \frac{1}{3}$ (B) $p = \frac{1}{2}$ (C) $p = \frac{2}{3}$ (D) true for all p

Solution. A, B are always independent by construction. The real constraint is $A \perp C$ (and by symmetry $B \perp C$). $P(C) = p^2 + (1-p)^2$, and $A \cap C = \{\text{both heads}\}$, so $P(A \cap C) = p^2$. Independence requires

$$p^2 = p \cdot [p^2 + (1-p)^2] \implies p = p^2 + (1-p)^2 \quad \text{(dividing by } p \neq 0\text{)}$$

$$p = 2p^2 - 2p + 1 \implies 2p^2 - 3p + 1 = 0 \implies (2p-1)(p-1) = 0.$$

So $p = \frac{1}{2}$ or $p = 1$ (degenerate, excluded since $p \in (0, 1)$). Only $\boxed{p = \frac{1}{2}}$ works.

Trap analysis: "True for all p" is the single most dangerous option here — the classic textbook version of this example always uses a fair coin, and it's easy to walk away believing the pairwise-independence property is robust to bias, when in fact it's a knife-edge property that fails for *every* $p \neq \frac{1}{2}$. $p = \frac{1}{3}, \frac{2}{3}$ are plausible-looking but don't satisfy the quadratic.

★ *Intuition:* Fairness isn't just a convenient simplification in the classic A, B, C example — it's load-bearing. The instant you bias one coin even slightly, "coin 1 is heads" starts leaking information about "the coins match" (because heads and tails are no longer symmetric outcomes), and the elegant pairwise-independent structure collapses.

Correct: (B) $p = \frac{1}{2}$

II. Bayesian Updating & Conditioning

Problem H4. *[Technique: Two-stage posterior-predictive (Bayes update, then LTP forward again)]*

Using the hidden-coin-type setup from your base sheet (fair p=.5 w.p. .5; head-biased p=.7 w.p. .3; tail-biased p=.3 w.p. .2; same coin flipped repeatedly): given the *first two* flips both came up heads, find $P(\text{3rd flip is heads})$.

Options: (A) 0.520 (B) 0.589 (C) 0.329 (D) 0.700

Solution. Step 1 (Bayes: update the type-posterior on HH). Reusing $P(HH) = 0.290$ from the base sheet:

$$P(\text{fair} \mid HH) = \frac{0.125}{0.290} \approx 0.431, \quad P(\text{hb} \mid HH) = \frac{0.147}{0.290} \approx 0.507,$$

$$P(\text{tb} \mid HH) = \frac{0.018}{0.290} \approx 0.062.$$

Step 2 (LTP: predict flip 3 using the updated posterior, not the prior).

$$P(\text{3rd}{=}H \mid HH) = 0.431(0.5) + 0.507(0.7) + 0.062(0.3) \approx \boxed{0.589}.$$

Trap analysis: 0.520 = used the *prior* marginal $P(H)$ and never updated on the evidence at all — ignoring two rounds of data. 0.329 = a denominator error: computed $P(HHH)/P(H)$ instead of $P(HHH)/P(HH)$, conditioning on the wrong event. 0.700 = incorrectly assumed the posterior fully concentrated on the head-biased type after seeing two heads.

★ *Intuition:* This is Bayesian updating in its most realistic form: you don't just get one piece of evidence and stop (as in the base sheet's single-flag problems) — you fold in new data, get a sharper belief about *which regime you're in*, and then predict forward using that sharper belief, not your original assumptions. It's exactly how a trader should update a view on "which volatility regime are we in" after each new data point, then re-price forward risk using the updated view rather than the stale prior.

▶ *Speed insight:* The predictive probability for the *next* observation is never just "look up a rate" — it's always (posterior-weighted average of the rate across every hypothesis still in play), which is why you can't skip the Bayes step even though the question ends by asking a plain LTP question.

Correct: (B) 0.589

Problem H5. *[Technique: Odds-form Bayes with a negative (disconfirming) result]*

Reusing Problem 13/14/15's monitor 1 (95% sens., 2% FPR) and monitor 2 (90% sens., 3% FPR) from your base sheet, and the same 1% base rate: monitor 1 flags *positive*, but monitor 2 flags *negative*. Find the posterior probability of a true anomaly.

Options: (A) 0.324 (B) 0.047 (C) 0.935 (D) 0.005

Solution. Prior odds $= \frac{1}{99}$. Monitor 1 (positive) contributes Bayes factor $\frac{0.95}{0.02} = 47.5$ as before. Monitor 2 is *negative*, so its Bayes factor uses the *miss* rate over the *true-negative* rate:

$$\text{BF}_2 = \frac{P(\text{neg} \mid \text{Anomaly})}{P(\text{neg} \mid \text{Normal})} = \frac{1-0.90}{1-0.03} = \frac{0.10}{0.97} \approx 0.1031.$$

$$\text{Posterior odds} = \frac{1}{99} \times 47.5 \times 0.1031 \approx 0.0495.$$

$$P(\text{Anomaly}) = \frac{0.0495}{1.0495} \approx \boxed{0.047}\ (4.7\%).$$

Trap analysis: 0.935 = used $\text{BF}_2 = \frac{0.90}{0.03} = 30$ instead — i.e. plugged in the *sensitivity* as if monitor 2 had also flagged positive, ignoring that a negative result should *pull the odds down*, not up. This is the single most dangerous mistake in this entire chapter: using a test's "accuracy numbers" without checking which direction the actual observed outcome points. 0.324 = only used monitor 1's evidence and forgot monitor 2 entirely. 0.005 is a plausible-looking overcorrection from a sign/reciprocal slip.

★ *Intuition:* Odds-form Bayes handles *disconfirming* evidence exactly the same mechanical way it handles confirming evidence — you just multiply by a Bayes factor *less than 1*, which shrinks your odds. A detective who gets one corroborating clue and one alibi shouldn't end up more confident than before; the arithmetic here enforces that automatically, but only if you correctly identify which conditional probability (the "hit" rate or the "miss" rate) the observed outcome calls for.

▶ *Speed insight:* Before writing down a Bayes factor, always ask: "did this test come back positive or negative?" A positive result's Bayes factor is (sensitivity)/(false-positive rate); a negative result's is (1−sensitivity)/(1−false-positive rate) — never the same fraction for both.

Correct: (B) 0.047

Problem H9. *[Technique: Bayes with correlated (non-independent) evidence — when the sequential-odds trick breaks]*

Same setup as Problem 15 on your base sheet, *except*: the two monitors share a common weakness (both over-fire during high-volume periods), so their false positives are correlated: $P(\text{both flag} \mid \text{Normal}) = 0.012$ (not $0.02 \times 0.03 = 0.0006$ as independence would predict). Sensitivities remain conditionally independent given a true anomaly: $P(\text{both flag} \mid \text{Anomaly}) = 0.95 \times 0.90 = 0.855$. Base rate 1%. Find the true $P(\text{Anomaly} \mid \text{both flag})$.

Options: (A) 0.935 (B) 0.419 (C) 0.324 (D) 0.855

Solution. Direct LTP/Bayes on the joint "both flag" event (odds-form doesn't cleanly apply here, since the two pieces of evidence are *not* conditionally independent given Normal):

$$P(\text{both flag}) = (0.855)(0.01) + (0.012)(0.99)$$

$$= 0.00855 + 0.01188 = 0.02043.$$

$$P(\text{Anomaly} \mid \text{both flag}) = \frac{0.00855}{0.02043} \approx \boxed{0.419}\ (41.9\%).$$

Trap analysis: 0.935 is *exactly* Problem 15's answer — it's what you get if you (wrongly) assume full conditional independence and multiply Bayes factors 47.5×30 as before. The correlated false-positive rate (0.012, twenty times larger than the independence-implied 0.0006) makes that assumption badly wrong here, and the true posterior is less than half of what naive sequential updating would claim. 0.324 = used only one monitor's evidence. 0.855 = reported the raw likelihood $P(\text{both flag} \mid \text{Anomaly})$ directly as if it were the posterior — a textbook confusion-of-the-inverse error.

★ *Intuition:* This is precisely the real-world failure mode behind "independent" risk models blowing up together: if two signals share a hidden common cause (here, high trading volume triggering both monitors' false-positive mechanisms), treating them as independent evidence massively overstates your confidence. The sequential-odds trick from Problem 15 is a fantastic shortcut *only* when the conditional-independence assumption is actually justified — always ask "could these two pieces of evidence share a common confounding cause?" before chaining Bayes factors.

▶ *Speed insight:* Whenever a problem explicitly gives you a joint conditional probability that differs from the product of the marginals (as $0.012 \neq 0.02 \times 0.03$ does here), that mismatch *is* the problem — it's telling you evidence is correlated and you must work with the joint likelihood directly, not decompose into separate Bayes factors.

Correct: (B) 0.419

Problem H10. *[Technique: Chain rule forward, Bayes inversion backward — full capstone]*

A three-trade sequence: $P(W_1) = 0.6$. $P(W_2 \mid W_1) = 0.7$, $P(W_2 \mid L_1) = 0.3$. Trade 3 depends on how many of the first two won: $P(W_3 \mid \text{both won}) = 0.8$; $P(W_3 \mid \text{exactly one won}) = 0.5$; $P(W_3 \mid \text{both lost}) = 0.2$. Given trade 3 won, find $P(\text{trade 1 also won})$.

Options: (A) 0.542 (B) 0.786 (C) 0.6 (D) 0.336

Solution. Build the full 8-path tree via the chain rule (probability of each path is the product along its branches):

$W_1W_2W_3$:	0.336	$L_1W_2W_3$:	0.060
$W_1W_2L_3$:	0.084	$L_1W_2L_3$:	0.060
$W_1L_2W_3$:	0.090	$L_1L_2W_3$:	0.056
$W_1L_2L_3$:	0.090	$L_1L_2L_3$:	0.224

(These sum to exactly 1.000 — a mandatory sanity check.) Now condition on W_3: sum every path ending in W_3:

$$P(W_3) = 0.336 + 0.090 + 0.060 + 0.056 = 0.542.$$

Of those, the ones that *also* start with W_1: $0.336 + 0.090 = 0.426$.

$$P(W_1 \mid W_3) = \frac{0.426}{0.542} \approx \boxed{0.786}.$$

Trap analysis: 0.542 = $P(W_3)$, the marginal — reported instead of the requested conditional, a confusion-of-the-inverse error at the

very end of an otherwise-correct computation. $0.6 = P(W_1)$, the *prior* win rate — this is what you'd (wrongly) answer if you ignored the evidence from trade 3 entirely. 0.336 is P(all three win), a natural but different quantity (the joint probability of one specific path, not the conditional probability being asked for).

★ *Intuition:* This is Bayes' theorem operating on an entire decision tree rather than a single test: knowing trade 3 won is like a "signal from the future" that should update your belief about what happened at trade 1, exactly the way knowing a court case was ultimately won might update your confidence that the initial filing was strong — even though trade 1's outcome causally happened *before* trade 3's, the information flows backward through the tree via Bayes' theorem exactly as cleanly as it flows forward via the chain rule.

▶ *Speed insight:* On any multi-stage chain-rule problem, always compute the full path table first (and verify it sums to 1!) before attempting any conditional question — trying to shortcut a "given the last stage, what happened at the first stage" question without the full tree is how the 0.542 vs. 0.786 confusion happens.

Correct: (B) 0.786

III. Classical Bayesian Traps

Problem H6. *[Technique: Monty Hall with a known, biased host preference]*

Same 3-door setup, but the host is known to prefer opening the higher-numbered door when he has a choice: if the car is behind your picked door (door 1), the host opens door 3 with probability $q = \frac{3}{4}$ and door 2 with probability $\frac{1}{4}$. He opens door 3. Find $P(\text{win} \mid \text{switch})$, *given this specific observation.*

Options: (A) $\frac{2}{3}$ (B) $\frac{4}{7}$ (C) $\frac{1}{2}$ (D) $\frac{3}{7}$

Solution. $P(\text{opens } 3 \mid \text{car} = 1) = \frac{3}{4}$, $P(\text{opens } 3 \mid \text{car} = 2) = 1$ (forced), $P(\text{opens } 3 \mid \text{car} = 3) = 0$.

$$P(\text{opens } 3) = \tfrac{1}{3}\left(\tfrac{3}{4}\right) + \tfrac{1}{3}(1) + \tfrac{1}{3}(0) = \tfrac{7}{12}.$$

$$P(\text{car} = 2 \mid \text{opens } 3) = \frac{\frac{1}{3}(1)}{\frac{7}{12}} = \boxed{\tfrac{4}{7}} \ (\approx 0.571).$$

Trap analysis: $\frac{2}{3}$ is the classic *unbiased*-host answer — reflexively applying it here ignores the fact that the host's known preference makes "opens door 3" *weaker* evidence against door 1 than in the symmetric case (a car-holding host with a strong preference for door 3 opens it often, so seeing door 3 opened is less surprising, hence less informative, than in the unbiased game). $\frac{1}{2}$ wrongly assumes the bias makes the observation completely uninformative. $\frac{3}{7}$ is the correct denominator but reports the "stay" probability instead of "switch."

★ *Intuition:* As a sanity check: averaged over which door the host actually opens (weighting by how often each door gets opened under this biased rule), switching still wins exactly $\frac{2}{3}$ overall — the host's bias doesn't change your *unconditional* advantage from switching, only how that advantage is distributed across "door 2 opened" ($\frac{4}{5}$ switch-win probability, since door-2-opens is a rarer, more surprising event under this host) versus "door 3 opened" ($\frac{4}{7}$). This is a genuinely subtle point: bias redistributes information across observations without changing the total.

▶ *Speed insight:* Whenever a Monty Hall variant tells you the host has a *known, asymmetric* rule for choosing between two losing doors, that rule changes $P(\text{observation} \mid \text{car} = \text{your door})$ away from the clean $\frac{1}{2}$ — always redo the full Bayes computation rather than reaching for $\frac{2}{3}$ from memory.

Correct: (B) $\frac{4}{7}$

Problem H7. *[Technique: Bertrand's Box with a non-uniform prior]*

Same three boxes (GG, GS, SS), but now box GG is selected with probability 0.5, box GS with probability 0.3, and box SS with probability 0.2 (not uniform). A random coin from the chosen box is gold. Find P(other coin is also gold).

Options: (A) $\frac{2}{3}$ (B) $\frac{10}{13}$ (C) $\frac{1}{2}$ (D) $\frac{5}{8}$

Solution.

$$P(\text{gold}) = (0.5)(1) + (0.3)(0.5) + (0.2)(0) = 0.5 + 0.15 = 0.65.$$

$$P(GG \mid \text{gold}) = \frac{(0.5)(1)}{0.65} = \frac{0.5}{0.65} = \boxed{\tfrac{10}{13}} \ (\approx 0.769).$$

Trap analysis: $\frac{2}{3}$ is the answer to the *original, uniform-prior* Bertrand's Box problem — it's burned into memory precisely because it's so famous, making it the most tempting wrong answer the instant the prior changes. $\frac{1}{2}$ is the naive "must be box GG or GS, so 50-50" answer, ignoring Bayes entirely. $\frac{5}{8}$ comes from using only the *prior* box probabilities ($\frac{0.5}{0.5+0.3}$) and forgetting to weight by each box's probability of actually producing a gold coin.

★ *Intuition:* This is exactly why memorized answers are dangerous in a timed exam: the mechanics of the problem (observe evidence, reweight your beliefs about which box you're in) are identical to the classic version, but plugging in the classic answer instead of redoing the two-line Bayes computation is a five-second shortcut that costs you the whole question.

Correct: (B) $\frac{10}{13}$

Problem H8. *[Technique: The "Tuesday Boy" extension of the Boy-or-Girl paradox]*

A family has two children; each child is independently a boy or girl (prob. $\frac{1}{2}$ each) and born on one of 7 days uniformly at random, independent of sex. Given "at least one child is a boy born on a Tuesday," find P(both children are boys).

Options: (A) $\frac{1}{3}$ (B) $\frac{13}{27}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Solution. Each child has 14 equally likely (sex, day) outcomes, so 196 equally likely outcomes for the pair. Let "BT" = "boy born Tuesday" (1 of the 14 per-child outcomes).

$$P(\text{neither child is BT}) = \left(\tfrac{13}{14}\right)^2 = \tfrac{169}{196}$$

$$\implies P(\text{at least one BT}) = \tfrac{27}{196}.$$

Among the 49 "both boys" outcomes (7×7), the ones with *no* Tuesday-boy number $6 \times 6 = 36$, so "both boys AND at least one BT" $= 49 - 36 = 13$ outcomes, i.e. probability $\frac{13}{196}$.

$$P(\text{both boys} \mid \text{at least one BT}) = \frac{13/196}{27/196} = \boxed{\tfrac{13}{27}} \ (\approx 0.481).$$

Trap analysis: $\frac{1}{3}$ is the answer to the *plain* "at least one boy" version (Problem 19(a) on your base sheet) — applying it here ignores that the extra "born on a Tuesday" qualifier, paradoxically, shifts the answer much closer to $\frac{1}{2}$. $\frac{1}{2}$ over-corrects, as if naming a specific day were equivalent to identifying one specific child (as in Problem 19(b)/(c)) — it isn't quite, which is exactly why the answer lands at $\frac{13}{27} \approx 0.481$, close to but not equal to $\frac{1}{2}$. $\frac{1}{4}$ is an unjustified naive guess treating the two conditions as roughly independent halvings.

★ *Intuition:* The Tuesday detail feels like it should be irrelevant "noise" — surely a boy's birth day can't change whether his sibling is a boy! And it doesn't, causally. But it changes *how rare and specific your conditioning event is*: "at least one boy" throws away only the GG outcomes, while "at least one boy born specifically on a Tuesday" throws away almost all of the boy-containing outcomes too (any family whose boy(s) happen to be born on the other 6 days), leaving a conditioning set that's disproportionately full of two-boy families (since a two-boy family gets two independent

chances to have hit "Tuesday," making it comparatively more likely to survive the filter). The rarer and more specific the identifying detail, the closer the answer creeps from $\frac{1}{3}$ toward $\frac{1}{2}$.

▶ *Speed insight:* This escalation pattern (add an ever-more-specific identifying detail; watch the answer drift from $\frac{1}{3}$ toward $\frac{1}{2}$) is a well-known family of interview questions — if you understand *why* it drifts (rarer details disproportionately favor the two-boy families surviving the filter), you can sanity-check any new variant without re-deriving from scratch.

Correct: (B) $\frac{13}{27}$

Calibration note: H1–H5, H7 are realistic 90–150s Stage-1 questions once the fusion is recognized. H6, H8, H9, and H10 are genuinely Stage-2/Finals-grade (each requires either a full enumeration or a two-stage Bayesian argument) — but every one of them is a direct escalation of a base-sheet problem, so if Sections I–II of the base 20-problem sheet are solid, none of these should feel unfamiliar, only longer.